

Blackjack Project

Pseudocode and Flowchart

One important note: The function `apply_action()` is not implemented yet. The flowchart shows where it belongs and clearly marks this limitation.

1. The Big Idea

The program takes information about a Blackjack hand, works out the value of the cards, checks which actions are allowed, and asks the player what they want to do.

Easy memory sentence: Input → Understand the hand → Calculate the value → Check the rules → Ask the player → Apply the action.

2. Pseudocode

Pseudocode is not real Python code. It is a simple English version of what the program should do.

START

1. Create a dictionary called `RANK_VALUES`.
Store the value of each card.
Example: 2 = 2, K = 10, A = 11.
2. Make a function called `HAND_VALUE(cards)`.
 - a. Add together the values of all the cards.
 - b. Count how many Aces are in the hand.
 - c. WHILE the total is above 21 AND there is an Ace:
 - Subtract 10 from the total.
 - This changes one Ace from 11 to 1.
 - d. RETURN the final total.
3. Make a function called `PARSE_STATE(text)`.
 - a. Receive text such as: A,A | 5 | N.
 - b. Split the text at the | symbol.
 - c. Separate the player's hand, dealer's face-up card, and flag.
 - d. Turn the hand into a list of cards.
 - e. Calculate the hand value.
 - f. RETURN all this information in one dictionary called state.
4. Make a function called `GENERATE_ACTIONS(state)`.
 - a. Start with HIT and STAND.
 - b. IF the player has exactly 2 cards, add DOUBLE.
 - c. IF the first two cards have the same rank, add SPLIT.
 - d. RETURN the list of allowed actions.
5. Make a function called `GET_USER_ACTION(state)`.
 - a. Find the allowed actions.
 - b. Show the player's hand and the dealer's face-up card.
 - c. Ask the player to choose an action.
 - d. Convert the answer to lower case and remove extra spaces.
 - e. IF the answer is in the allowed actions, RETURN it.
 - f. OTHERWISE, show an error and ask again.
6. Make a function called `APPLY_ACTION(state, action, next_card)`.
This should perform the chosen action.

In the current code, this function is not implemented yet.

7. Test the program using: A,A | 5 | N.

Print the parsed state and the allowed actions.

END

3. Flowchart

Follow the arrows from START to END. The diamonds are decision points: the program asks a question and follows a different path depending on YES or NO.

